

ECE 501 DIGITAL IMAGE PROCESSING

SAR–Optical Feature Fusion for Crop-Type Classification REPORT — WEEK 1 PROJECT 19 — GROUP 20

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1) Introduction and Problem Statement

Crop-type mapping and yield estimation from satellite remote sensing are among the most widely used applications of agricultural monitoring today. Optical sensors such as Sentinel-2 capture spectral reflectance, which is directly tied to plant pigment and canopy health, whereas SAR sensors such as Sentinel-1 measure the radar backscatter returned from the crop canopy, which instead reflects structural and moisture-related properties of the vegetation and soil. Unlike optical imaging, SAR does not depend on daylight or clear skies, since radar signals penetrate cloud cover. Each modality, however, also carries its own limitations: optical imagery is frequently unavailable during key stages of the growing season due to cloud cover, while SAR imagery tends to be noisier and its backscatter is less directly interpretable in terms of biophysical crop variables.

This project, titled "SAR–Optical Feature Fusion," addresses this trade-off by combining classical, hand-crafted features extracted independently from SAR and optical imagery of the same agricultural fields, with the goal of classifying crop type and, where available, relating this to per-pixel yield information. Three classical machine-learning pipelines will be built — one using optical features only, one using SAR features only, and one using the fused feature set — and their performance will be compared using classification accuracy, F1-score, and confusion matrices to quantify any improvement gained through fusion.

The core tasks for this project are:

- (i) extracting classical hand-crafted features separately from Sentinel-2 optical data and Sentinel-1 SAR data for the same fields, along with their associated crop-type labels and per-pixel yield masks where provided;
- (ii) fusing the optical and SAR feature vectors at the feature level; and

(iii) training and evaluating classical ML models (e.g., Random Forest, Support Vector Machine) on the optical-only, SAR-only, and fused feature sets, reporting and comparing their accuracy, F1-score, and confusion matrices.

2) Reasons for Choosing This Topic

Our group selected this topic for several reasons tied directly to the goals of a Digital Image Processing course project. First, it lets us work with two fundamentally different imaging mechanisms — passive multispectral optical imaging and active coherent radar imaging — within a single pipeline, giving us hands-on exposure to a range of classical DIP techniques: spectral index computation, GLCM-based texture analysis, speckle filtering, and feature normalization, all applied to real, heterogeneous satellite data rather than a single modality or synthetic imagery.

Second, the underlying problem is practically motivated. Optical data availability drops sharply during cloudy periods of the growing season, which can be a real constraint depending on region and climate. SAR, being weather-independent, offers a way to fill these gaps — and evaluating how much SAR can genuinely compensate for missing optical information is a meaningful and current question in agricultural remote sensing. Structuring our comparison at the feature level (separately evaluating optical-only, SAR-only, and fused settings) lets us isolate and directly measure this benefit rather than assuming it.

Third, this project intentionally avoids deep learning in favor of hand-crafted features and classical shallow learning models. This fits the introductory nature of the course well — it requires us to understand the physical meaning behind spectral indices, GLCM texture statistics, and SAR backscatter behavior, rather than relying on an opaque learned representation. It also keeps the project scope realistic for a team our size within a single semester.

Finally, access to a purpose-built, recent benchmark dataset (YieldSAT) covering the specific countries, crops, and label types relevant to this problem made the project feasible within our timeline, while still leaving room to explore our own SAR data acquisition and feature-fusion strategy.

3) Brief Background

SAR–optical fusion for agricultural mapping is an active area of remote-sensing research. Prior work has shown that combining Sentinel-1 and Sentinel-2 data for crop-type classification and phenology monitoring generally outperforms either modality alone, particularly in regions with frequent cloud cover. Classical machine learning models — Random Forest and Gradient Boosting in particular — remain competitive baselines for crop-type mapping tasks when paired with well-engineered spectral and radar-backscatter features, without requiring deep learning.

For our assigned country pair, Germany and Uruguay, this framing is directly relevant: Germany's temperate climate and well-studied agricultural regions have been the subject of several Sentinel-1/Sentinel-2 fusion studies for crop-type mapping in European contexts, giving

us a reasonably strong base of prior work to draw on for feature choices and expected performance benchmarks. Uruguay, while less extensively studied in this specific fusion context, shares similar major crop types (soybean, wheat, corn) with other South American Sentinel-1 crop-mapping studies, which we plan to draw on for comparison once we complete a fuller literature review next week.

This motivates the classical (non-deep-learning) feature set we plan to extract: optical vegetation indices (e.g., NDVI, NDRE, NDMI), SAR backscatter statistics (VV, VH, VV/VH ratio) with GLCM texture, and a concatenated fused feature vector combining both. Since Sentinel-1 SAR data is not included in the provided YieldSAT package, we will acquire it ourselves via Google Earth Engine (GEE) for the same fields as the provided optical data, following this general approach:

- Extract field boundary polygons/centroid coordinates and acquisition dates from the YieldSAT optical metadata to define the area and time window for SAR retrieval.
- Pull Sentinel-1 GRD data in Interferometric Wide (IW) swath mode (VV and VH polarizations), matched to the growing-season dates of the corresponding optical/yield samples.
- Apply standard GEE preprocessing for Sentinel-1 GRD: border noise removal, thermal noise removal, radiometric calibration to backscatter coefficient σ^0 , and terrain correction — with an optional speckle-reduction filter (e.g., Lee or Refined Lee) before feature extraction.
- Export each SAR patch in the same geometry as its corresponding optical patch, so both are spatially co-registered for fusion.

4) Dataset Identification

The dataset provided for this project is drawn from YieldSAT, a recently published multimodal benchmark for high-resolution crop yield prediction (Miranda et al., CVPR 2026; project page: yieldsat.github.io). It provides Sentinel-2 optical imagery, crop-type labels, and per-pixel yield masks for two of four available countries — our group is working with **Germany and Uruguay**.

Key dataset attributes relevant to our project:

- Coverage spans four countries in total (Germany, Uruguay, Argentina, Brazil) across roughly 138,288 hectares of selected fields.
- Crop types include corn (maize), rapeseed, soybean, and wheat.
- Yield labels are derived from over 12.2 million GPS-tagged, per-pixel yield-monitor readings (wet yield, moisture content, timestamp), rasterized to match satellite resolution (10 m).
- Optical imagery consists of 113,555 time-series Sentinel-2 patches, each labeled with crop type and paired with a yield mask. Additional metadata (weather, soil, topography) is available but outside our project's scope — we use only the optical bands, crop-type labels, and yield masks.

Since YieldSAT is fundamentally a yield-regression dataset, we will derive our classification target directly from the crop-type attribute (corn / rapeseed / soybean / wheat) associated with each field, which for our country pair (Germany + Uruguay) forms a multi-class classification problem across the crop types present in both countries.

5) Planned Methodology

- **Data preparation:** Download and organize the optical imagery, crop-type labels, and yield masks provided for Germany and Uruguay from YieldSAT.
- **SAR acquisition and co-registration:** Retrieve matching Sentinel-1 SAR data via Google Earth Engine for the same fields and time windows, and co-register/clip it to align spatially with the optical patches.
- **Optical feature extraction:** For each field patch, compute vegetation indices (NDVI, NDRE, NDMI, EVI) and GLCM texture features (contrast, homogeneity, energy, correlation).
- **SAR feature extraction:** Compute backscatter statistics (mean and variance of VV, mean and variance of VH, VV/VH ratio) and GLCM texture features for each SAR patch.
- **Feature fusion:** Normalize and concatenate the optical and SAR feature vectors per field to form the fused feature representation.
- **Model training and evaluation:** Train classical ML classifiers (Random Forest, SVM) separately on optical-only, SAR-only, and fused features, using a consistent train-test split and cross-validation setup across all three settings for a fair comparison, and report accuracy, F1-score, and confusion matrices for each.

6) Work Completed This Week and Plan Ahead

This being our Week 1 submission, our group has focused on understanding the problem statement, reviewing background context on SAR–optical fusion for agriculture, and confirming our dataset and toolset (YieldSAT for optical/labels, Google Earth Engine for SAR acquisition). We selected Germany and Uruguay as our country pair.

Alongside this, we have already set up our project's GitHub repository ([DIP_2026_Project19-SOFF_Group20](#)) with the required directory structure (Mid_Sem_Report, End_Sem_Report, Results, Codes), an initial project plan and timeline, and a starter code scaffold for the pipeline stages (data loading, optical feature extraction, SAR feature extraction, fusion, and model training/evaluation) that we will build out with real implementation in the coming weeks. No feature extraction or model training has been run on actual data yet.

For next week, we plan to complete a fuller literature review specifically on Sentinel-1/Sentinel-2 fusion studies relevant to Germany and Uruguay, begin implementing the Google Earth Engine-based Sentinel-1 acquisition pipeline, and start adapting our optical feature extraction code to the real YieldSAT data format.

7) References

- [1] A. Miranda et al., "YieldSAT: A Multimodal Benchmark Dataset for High-Resolution Crop Yield Prediction," CVPR 2026. Project page: <https://yieldsat.github.io/>
- [2] Google Earth Engine, "Sentinel-1 SAR GRD: C-band Synthetic Aperture Radar Ground Range Detected," Data Catalog documentation, Google LLC.
- [3] N. Gorelick et al., "Google Earth Engine: Planetary-scale geospatial analysis for everyone," *Remote Sensing of Environment*, 2017.
- [4] Studies on Sentinel-1/Sentinel-2 fusion for crop-type mapping in Germany and South America (to be expanded with full citations during Week 2 literature review).